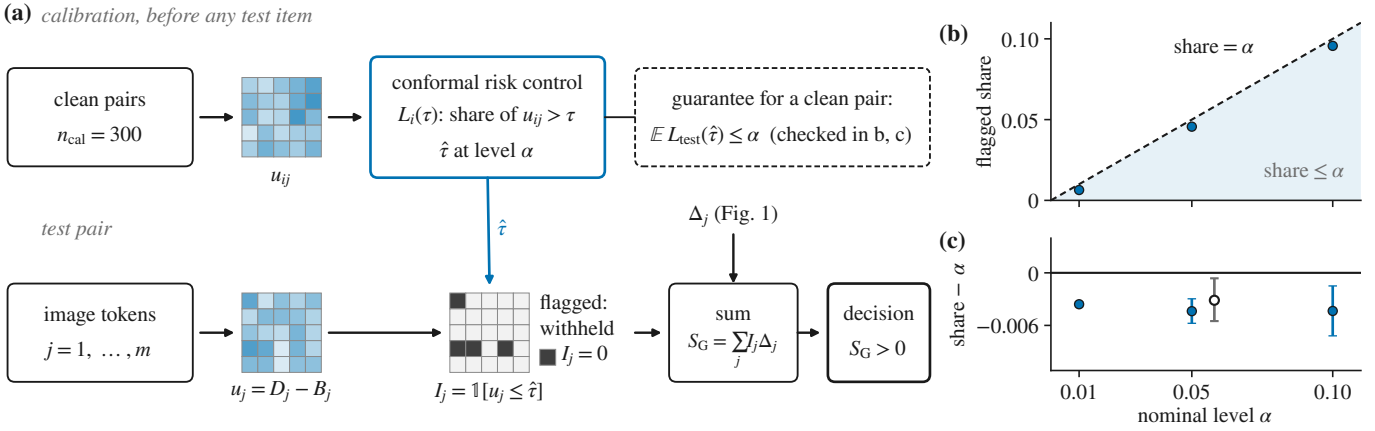
(a) calibration, before any test item
clean pairs
$n_{cal} = 300$
u_{ij}
conformal risk control
$L_i(\tau)$: share of $u_{ij} > \tau$
$\hat{\tau}$ at level α
guarantee for a clean pair:
$\mathbb{E}\, L_{test}(\hat{\tau}) \le \alpha$ (checked in b, c)
$\hat{\tau}$
Δ_j (Fig. 1)
test pair
image tokens
$j = 1, \ldots, m$
$u_j = D_j - B_j$
$I_j = \mathbb{1}[u_j \le \hat{\tau}]$
flagged: withheld
$I_j = 0$
sum
$S_G = \sum_j I_j \Delta_j$
decision
$S_G > 0$
(b)
flagged share
0.10
0.05
0
share = α
share ≤ α
(c)
share − α
0
−0.006
0.01
0.05
0.10
nominal level α